

FUTENMA MCAS, Japan

WMO: 479330

Lat: 26.274N

Lon: 127.756E

Elev: 257

StdP: 14.56

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 42215

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	51.8	53.4	37.1	32.7	55.2	38.9	35.1	56.0	25.2	56.8	23.2	58.5	10.1	10	0.427

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.4	90.3	79.9	89.5	79.8	88.2	79.5	82.6	86.6	81.7	85.9	80.9	85.3	11.8	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.2	163.7	85.0	80.5	159.6	84.3	79.4	154.0	83.8	46.8	87.0	45.7	86.2	45.0	85.7	87.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.0	21.4	19.0														
			DB	48.7	93.2	2.3	2.3	47.0	94.9	45.7	96.2	44.5	97.5	42.8	99.1	
			WB	43.1	84.8	2.0	1.5	41.7	85.8	40.5	86.7	39.4	87.5	38.0	88.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	73.4	62.2	62.4	65.3	70.1	75.1	80.8	84.4	84.1	81.8	77.1	71.5	65.1
	DBStd	8.87	4.39	4.83	5.00	4.72	3.75	3.68	1.97	1.93	2.55	3.39	3.78	4.30
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	336	110	101	60	10	0	0	0	0	0	0	2	53
	CDD50	8541	379	348	474	604	779	925	1065	1058	954	841	645	469
	CDD65	3402	24	29	69	164	314	475	600	593	504	376	197	57
	CDH74	30063	8	10	79	479	1722	4751	7360	7147	5306	2583	576	42
	CDH80	9463	0	0	1	58	239	1406	2981	2777	1566	412	23	0
Wind	WSAvg	9.9	9.4	9.8	10.1	10.0	9.5	10.9	10.2	10.1	9.9	9.8	9.5	9.6
Precipitation	PrecAvg	81.8	4.8	4.3	6.0	6.5	9.6	10.1	7.5	9.8	7.1	6.3	4.3	5.5
	PrecMax	123.4	12.2	16.0	14.4	18.1	27.2	22.4	24.4	25.6	18.1	23.0	10.2	32.4
	PrecMin	40.7	1.1	1.1	0.8	0.8	0.6	0.9	0.5	1.5	1.7	0.2	0.2	0.9
	PrecStd	21.4	2.2	3.3	3.5	3.9	5.5	5.5	6.7	5.8	3.8	4.8	3.0	5.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.2	75.4	79.0	84.6	86.8	90.7	92.3	91.9	90.2	87.6	82.5	77.3
		MCWB	68.2	69.2	71.2	75.8	78.3	80.6	80.2	80.2	79.5	78.1	75.1	70.5
	2%	DB	73.0	73.3	76.2	81.0	84.5	89.2	90.5	90.0	88.2	85.0	80.4	75.2
		MCWB	66.7	67.8	70.0	74.5	77.1	80.0	79.8	80.1	78.9	76.9	73.5	68.5
	5%	DB	70.6	71.5	73.9	78.8	82.6	87.8	89.6	89.3	87.6	83.9	78.8	73.5
		MCWB	64.3	66.3	68.5	72.6	76.4	79.4	79.8	80.0	78.8	76.2	71.8	66.8
	10%	DB	69.4	69.9	72.0	76.6	80.7	86.2	88.2	87.9	86.1	82.4	77.2	71.7
		MCWB	63.1	65.1	67.2	71.5	75.2	79.0	79.6	79.7	78.2	75.2	70.9	65.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.7	71.8	73.3	77.5	80.3	83.6	84.5	83.6	82.4	80.7	77.0	72.8
		MCDB	73.1	73.9	76.1	83.1	83.4	87.2	88.4	87.8	86.2	83.4	80.0	75.2
	2%	WB	68.0	69.6	71.4	75.6	78.8	82.0	82.6	82.0	81.2	79.2	75.1	70.7
		MCDB	71.6	71.9	74.4	79.6	82.6	85.6	86.9	86.5	85.2	82.5	78.5	73.4
	5%	WB	65.8	67.6	70.1	74.1	77.6	80.9	81.6	81.2	80.4	78.1	73.8	68.6
		MCDB	69.3	70.3	73.1	77.6	81.2	84.9	86.1	85.9	84.5	81.7	77.1	72.0
	10%	WB	64.1	65.8	68.4	72.7	76.4	80.0	80.8	80.6	79.6	76.9	72.6	66.9
		MCDB	67.9	69.0	71.8	75.7	80.0	84.1	85.5	85.4	83.9	81.0	76.1	70.9
Mean Daily Temperature Range	5% DB	MDBR	8.9	8.9	9.6	9.3	8.7	8.1	8.4	8.1	8.4	8.8	9.0	9.0
		MCDBR	10.2	9.6	10.3	10.3	9.6	9.0	9.4	9.4	9.4	9.5	9.4	9.9
		MCWBR	7.8	7.7	7.8	5.9	5.7	4.4	4.5	4.4	4.7	5.1	6.1	6.8
	5% WB	MCDBR	8.9	8.6	9.7	9.2	9.0	8.5	8.6	8.3	8.1	8.1	8.2	8.4
		MCWBR	8.2	8.2	8.7	6.6	6.6	5.2	5.3	5.0	5.0	5.3	6.3	7.5
Clear-Sky Solar Irradiance	taub	0.455	0.484	0.557	0.565	0.509	0.469	0.469	0.487	0.482	0.480	0.463	0.454	
	taud	2.138	2.076	1.905	1.946	2.169	2.350	2.343	2.281	2.264	2.209	2.213	2.147	
	Ebn at Noon	247	250	239	241	254	263	262	258	256	249	244	241	
	Edh at Noon	43	49	61	60	48	40	40	43	43	43	40	41	
All-Sky Solar Radiation	RadAvg	771	974	1202	1403	1563	1729	2066	1889	1657	1302	952	778	
	RadStd	92	146	114	139	203	157	167	128	124	113	88	76	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	+0.74	+0.95	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page